

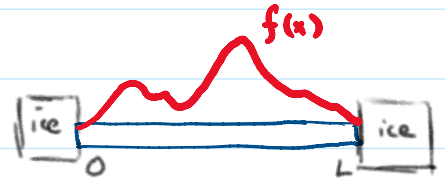
Solving the 1D Heat Equation (Filled In)

Sunday, July 13, 2025 1:25 AM

4. The Heat Equation:

4.1 The problem:

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$



BC: $u(0, t) = 0$, $u(L, t) = 0$ for all $t \geq 0$

IC: $u(x, 0) = f(x)$

As a class, we are going to solve this DE together.

4.2 Separation of variables: $u(x, t) = F(x)G(t)$

Derive two ODEs from the PDE.

$$\begin{aligned} \frac{\partial}{\partial t} F(x)G(t) &= c^2 \frac{\partial^2}{\partial x^2} F(x)G(t) \Rightarrow F(x) \dot{G} = c^2 G(t) F''(x) \\ \Rightarrow \dot{G}/c^2 G &= F''/F = k \end{aligned}$$

$$\Rightarrow \dot{G} - kc^2 G = 0 \quad \& \quad F'' - kF = 0$$

Using BC on F : must have $k < 0 \Rightarrow k = -p^2$, so the 2 equations are:

$$F'' + p^2 F = 0$$

$$G + c^2 p^2 G = 0$$

4.3 One of the ODEs: $F'' + p^2 F = 0$

What does the solution to this look like?

$F'' + p^2 F = 0$ is a Simple Harmonic Oscillator

$$\Rightarrow F(x) = A \cos(px) + B \sin(px)$$

using $F(0) = 0$ get $A = 0$, so $F(x) = B \sin(px)$

using $F(L) = 0$ get $PL = n\pi \Rightarrow p = n\pi/L$

Thus $F(x) = B \sin\left(\frac{n\pi}{L}x\right)$

4.4 The other ODE: $\dot{G} + c^2 p^2 G = 0$
 since $p = \frac{n\pi}{L} \Rightarrow \dot{G} + \left(\frac{n\pi c}{L}\right)^2 G = 0$

What does the solution look like?

Get $\frac{\dot{G}}{G} = -\left(\frac{n\pi c}{L}\right)^2 \Rightarrow \ln(G) = -\left(\frac{n\pi c}{L}\right)^2 t$

$\therefore G(t) = \alpha e^{-\left(\frac{n\pi c}{L}\right)^2 t}$

Thus $u(x, t) = F(x)G(t) = B \sin\left(\frac{n\pi}{L}x\right) \alpha e^{-\left(\frac{n\pi c}{L}\right)^2 t} \Rightarrow B_n \sin\left(\frac{n\pi x}{L}\right) e^{-\left(\frac{n\pi c}{L}\right)^2 t}$

4.5 Overall solution for a given n :
 $u_n(x, t) = B_n \sin\left(\frac{n\pi x}{L}\right) e^{-\left(\frac{n\pi c}{L}\right)^2 t}$

Then, using Fourier Series, the complete solution is:

$$u(x, t) = \sum_{n=1}^{\infty} u_n(x, t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-\left(\frac{n\pi c}{L}\right)^2 t}$$

4.6 How do we use the initial condition: $u(x, 0) = f(x)$?

As a Fourier Series:

$$\sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-\left(\frac{n\pi c}{L}\right)^2 (0)} = f(x)$$

$$\Rightarrow B_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

4.7 Some easy cases: For the heat equation: $u_t = c^2 u_{xx}$, $c \approx 1.158 \text{ cm}^2/\text{s}$

$$u(0, t) = 0 = u(80, t) \quad (\text{i.e., } L=80)$$

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$$u(x, 0) = f(x)$$

$$(i) \quad f(x) = 100 \sin\left(\frac{\pi}{80} x\right)$$

$$\text{We know: } u(x, t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi}{80} x\right) e^{-\left(\frac{n\pi}{80}\right)^2 t}$$

$$\text{where } B_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{80} x\right) dx = \frac{2}{L} \int_0^L 100 \sin\left(\frac{\pi}{80} x\right) \sin\left(\frac{n\pi}{80} x\right) dx = \begin{cases} 100 & \text{if } n=1 \\ 0 & \text{otherwise} \end{cases}$$

$$\Rightarrow u(x, t) = 100 \sin\left(\frac{\pi x}{80}\right) e^{-\left(\frac{\pi}{80}\right)^2 t} \approx 100 \sin\left(\frac{\pi x}{80}\right) e^{-0.001785 t}$$

$$(ii) \quad f(x) = 100 \sin\left(\frac{3\pi x}{80}\right)$$

$$\text{So } B_n = \begin{cases} 100 & \text{if } n=3 \\ 0 & \text{otherwise} \end{cases} \Rightarrow u(x, t) = 100 \sin\left(\frac{3\pi}{80} x\right) e^{-0.01607 t}$$

4.8 What would be a physical difference between these two scenarios?

e.g. starting with 100°C , how long does it take for the temperature to fall to 50°C ?

$$\text{For (i): } u(x, t) = 100 \sin\left(\frac{\pi}{80} x\right) e^{-\left(\frac{\pi}{80}\right)^2 t}$$

$$u_{\max} = 100 e^{-\left(\frac{\pi}{80}\right)^2 t}$$

$$\text{want } u_{\max} = 50 \Rightarrow 100 e^{-\left(\frac{\pi}{80}\right)^2 t} = 50$$

$$\Rightarrow e^{-\left(\frac{\pi}{80}\right)^2 t} = \frac{1}{2}$$

$$\Rightarrow -\left(\frac{\pi}{80}\right)^2 t = \ln \frac{1}{2}$$

$$\Rightarrow -0.001785 t = -0.693$$

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$$\Rightarrow t = 388 \text{ sec} \approx 6.5 \text{ min}$$

$$\text{For (ii)} \quad u(x,t) = 100 \sin\left(\frac{3\pi x}{80}\right) e^{-\left(\frac{3\pi c}{80}\right)^2 t}$$

This results in:

$$-\left(3 \frac{\pi c}{80}\right)^2 t = \ln\left(\frac{1}{2}\right)$$

$$\Rightarrow 9\left(-0.001785\right)t = -0.693$$

$$\Rightarrow -0.01607t = -0.693$$

$$\Rightarrow t \approx 43 \text{ sec} \quad [9 \text{ times as fast as (i)!}]$$

Physically: The higher modes die off faster by an order of n^2 .

For large values of t :

$$u(x,t) \sim \left[\frac{2}{L} \int_0^L f(x) \sin\left(\frac{\pi x}{L}\right) dx \right] \sin\left(\frac{\pi x}{L}\right) e^{-\left(\frac{\pi c}{L}\right)^2 t}$$

4.9 Limitations:

- What if the bar is not placed in iced baths at the ends?
- What if the bar and the ends are insulated?
- What if the bar is of infinite length?